

UNIT 02

CHEMISTRY OF OUTER- TRANSITION ELEMENTS



Outer transition elements:

“Elements which possess partially filled d-orbitals in atomic or ionic state.”

- The elements of group III-B to II-B are called Non-typical transition elements.
- The elements of I-B to VIII-B are called typical transition elements (Except III-B & II-B).
- The general valence shell electronic configuration is $ns^{1-2}, (n-1)d^{1-10}$

Give reasons for the following:

- Zn, Cd and Hg are not regarded as transition metal.
- d-block elements exhibit a large number of oxidation state than f block elements.

(i) Group II B elements i.e. Zn, Cd, Hg have completely filled $(n-1)$ d-orbital in atomic & ionic state & thus these elements are not considered as transition elements.

(ii) The difference in the energy of $(n-1)$ d orbital and ns orbital is very small and thus both subshells are used for bond formation. Whereas in f-block elements $(n-2)$ f orbitals lie underneath and hence are not available for bond formation.

General Features of Transition Elements:

1. They all are metallic in nature.
2. Some of the transition elements play an important role in the industry. These metals are Ti, Cr, Fe, Ni, Cu, Mo, W, Zr, Nb, Ta and Th etc.
3. They are all hard and strong metal with high melting and boiling points. They are good conductors of heat and electricity.
4. They form alloys with one another and other elements of periodic table as well.
5. Their ions and compounds are colored in the solid state and the solution state.

Exceptional electronic configuration of Cr & Cu:

- Cr ($z=24$) has exceptional electronic configuration that is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$ instead of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$.
- Because half-filled electronic configuration is more stable than partial filled electronic configuration. Therefore, in order to attain stability electron of 4s orbital jumps to 3d orbital.
- Cu ($z=29$) has exceptional electronic configuration that is $[Ar] 4s^1 3d^{10}$ instead of $[Ar] 4s^2 3d^9$.
- Because fulfilled & half-filled electronic configuration are more stable than partial filled electronic configuration. Therefore, in order to attain stability electron of 4s orbital jumps to 3d orbital.

The first-row transition metals.



Scandium (Sc)



Titanium (Ti)



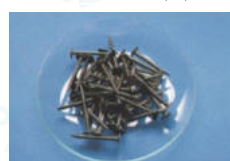
Vanadium (V)



Chromium (Cr)



Manganese (Mn)



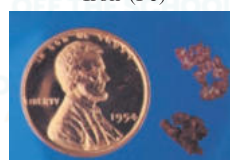
Iron (Fe)



Cobalt (Co)



Nickel (Ni)



Copper (Cu)

General Properties:**1) Binding Energy:**

"The amount of energy required to separate the components of atoms, nuclei or particles."

- Decreases down the group.
- Increases along the period.

REASON:

Why do the d-block elements show different oxidation states?

2) Variable Oxidation States:

- Transition elements are electropositive, so they have positive oxidation states.
- All 3d series elements show an oxidation state of +2 in addition to higher oxidation states when the electrons of 4s-orbital take part in bonding.

Reason for showing variable oxidation states:

They show variable oxidation states. The reason is that they have d-electrons in addition to s-electrons for the purpose of bond formation. These elements have several (n-1) d and ns electrons. The energies of (n-1) d and ns orbitals are very close to each other. The (n-1) d electrons are as easily lost as ns electrons. In the highest oxidation states of first five elements, all s and d-electrons are used for bonding.

Oxidation states of the first-row transition metals. The most stable oxidation numbers are shown in color. The zero oxidation state is encountered in some compounds, such as Ni(CO)_4 and Fe(CO)_5 .

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
				+7				
			+6	+6	+6			
		+5	+5	+5	+5			
	+4	+4	+4	+4	+4	+4		
+3	+3	+3	+3	+3	+3	+3	+3	+3
	+2	+2	+2	+2	+2	+2	+2	+2
								+1

3) Catalytic Activity:

Most of the transition elements and their compounds are used as catalysts.

- The reason is that the transition metals show variety of oxidation states. In this way, they can form intermediate products with various reactants.
- They also form interstitial compounds which can absorb an activator to the reacting species.

Examples:

Reactions	Catalyst
i. Oxidation of SO_2 & SO_3 .	V_2O_5
ii. Addition reaction in Alkene.	Ni, Pt & Pd
iii. Synthesis of Ammonia.	Fe

17

Electron Configurations and Other Properties of the First-Row Transition Metals

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
Electron configuration									
M	$4s^2 3d^1$	$4s^2 3d^2$	$4s^2 3d^3$	$4s^1 3d^5$	$4s^2 3d^5$	$4s^2 3d^6$	$4s^2 3d^7$	$4s^2 3d^8$	$4s^1 3d^{10}$

4) Magnetic Behavior:

Many transition elements and their compounds are paramagnetic.

• **The compounds attracted by the magnetic field are called paramagnetic.**

Paramagnetism is due to the unpaired electrons present in the metals and their compounds. Examples Mn^{2+} , Fe^{2+} etc.

• **The substances which can be magnetized are called ferromagnetic.**

For example, Fe, Co and Ni are ferromagnetic.

• **Some substances in which even number of electrons are present, and have paired spins are diamagnetic and slightly repelled by magnetic field.**

For example, Zn^{+2} , Sc^{+3} etc.

5) Alloy Formation:

Alloy is a homogenous mixture of two or more than two metals. Transition metals form alloys with each other.

Reason:

- Transition elements have almost similar sizes and atoms of the one metal can easily take up positions in crystal lattice of others and form interstitial alloys.
- They form substitution alloys among themselves.

Example:

- Alloy steels are the materials in which the iron atoms are substituted by Cr, Mn and Ni. Steel has more useful properties than iron.
- Brass, bronze and coinage alloys are the best alloys.

**Properties of alloys:**

As alloys are prepared according to the requirements, their characteristics are different, yet few properties are common which are as follows:

1. Alloys are comparatively cheap.
2. They are strong and flexible but hard alloys can also be prepared.
3. They have long life because they do not corrode.
4. They are durable.
5. They are better conductor but non-conductor alloys are also prepared.
6. They have high melting points.

Why alloys are harder as compared to their constituent metals?

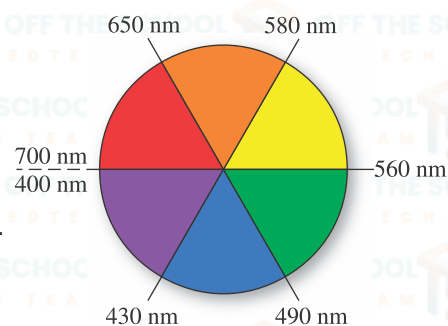
Ans. In the solid state, a pure metal has a giant metallic structure. The atoms are arranged in layers. When a force is applied, the layers may slide over each other. The greater the force needed, the harder and stronger the metal. In a pure metal, the force needed to make the layers slide over each other is small. This explains why many pure metals are soft.

In an alloy, there are atoms of different sizes. The smaller or bigger atoms distort the layers of atoms in the pure metal. This means that a greater force is required for the layers to slide over each other. The alloy is harder and stronger than the pure metal.



6) Color of Complexes:

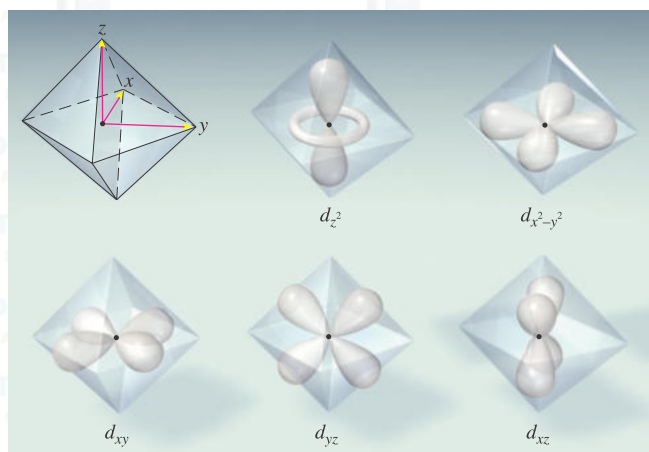
- When white light is allowed to fall on a complex, the following things may occur:
- The complex may absorb the whole white light. In this case it appears black.
- The complex may absorb some of it and may reflect or transmit the light. In this case the complex has some color. i.e. it is colored.
- Color is produced in the substance due to absorption and emission of the photon, in visible region of the spectrum.



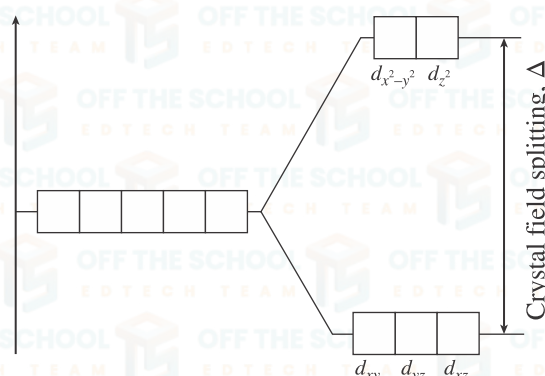
A color wheel with appropriate wavelengths. Complementary colors, such as red and green, are on opposite sides of the wheel.

Crystal Field Theory:

Transition elements can form colored compounds which can be explained by Crystal Field Theory (CFT). According to this theory, when ligands coordinate to a metal ion, the energies of the d-orbitals are no longer the same, d-orbitals split into two parts i.e; t_{2g} and e_g orbitals. t_{2g} has lower energy than e_g orbitals. The difference in energies between t_{2g} and e_g orbital is Δ_o .

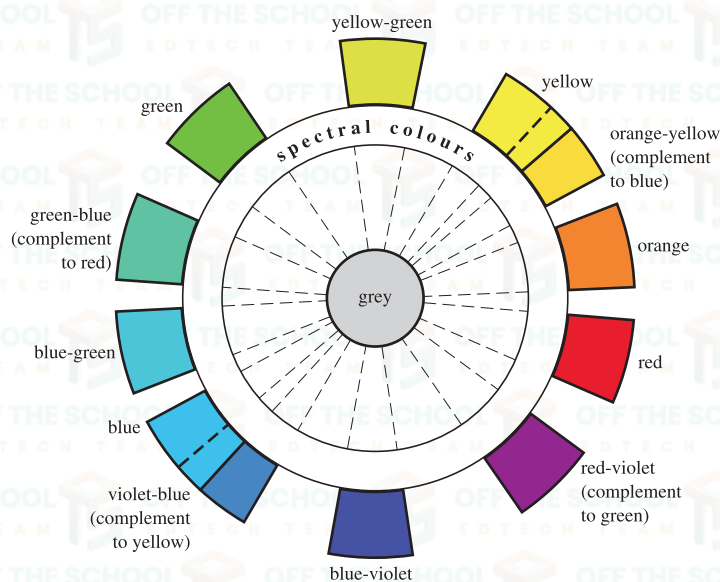


The five d orbitals in an octahedral environment. The metal atom (or ion) is at the center of the octahedron, and the six lone pairs on the donor atoms of the ligands are at the corners.



Crystal field splitting between d orbitals in an octahedral complex.

A colour wheel. Colours opposite each other are called complementary colours. When a substance absorbs light of a particular colour, the light that is reflected or transmitted has the colour of its complement. Thus, something that absorbs red light appears green-blue, and vice versa.

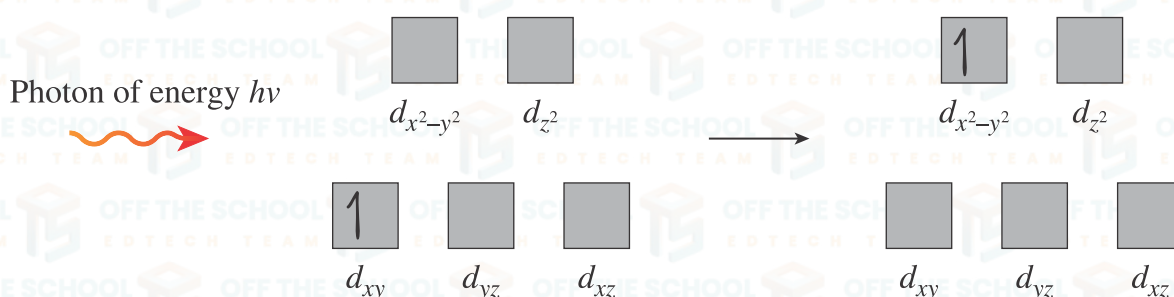


Explanation:

When transition elements make the compounds, then the electron of d-orbital are involved in bonding. For this reason, d-orbital are splitted into two levels. One group of d-orbital get little higher energy. The electrons present in lower d-orbital absorb the photons of energy and are promoted to higher level. The photon, which is involved for the jumping of electron, is responsible for showing definite color. Basically, unabsorbed light is transmitted and it decides the nature of color. This transition is called (d-d) transition.

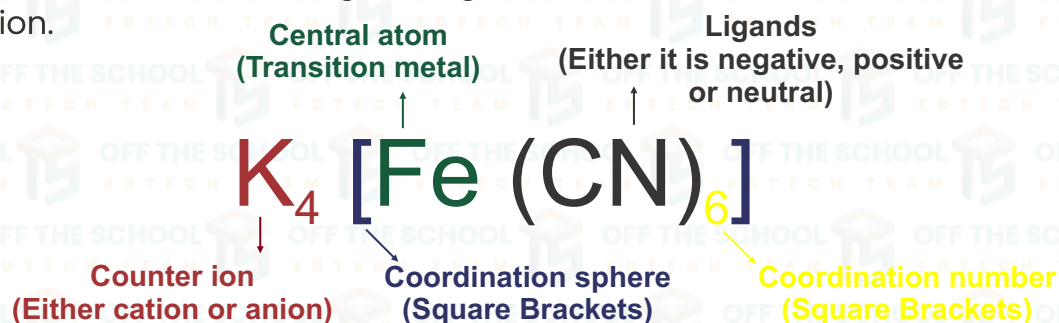


Colors of some of the first-row transition metal ions in solution. From left to right: Ti^{3+} , Cr^{3+} , Mn^{2+} , Fe^{3+} , Co^{2+} , Ni^{2+} , Cu^{2+} . The Sc^{3+} and V^{5+} ions are colorless.

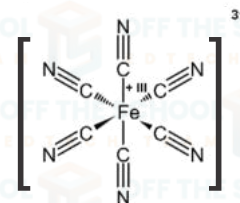


7) Formation of Coordination Complex:

- Transition metals form complex compounds.
- It is due to the presence of vacant d-orbitals as well as they show variable oxidation state.
- They can easily accept and donate the electron pairs.
- They are of small size and high charged this is also a reason behind the complex formation.



✓ **Coordination Complex:** The compounds in which central transition metals are surrounded by certain negative or neutral ligands. As you see the $[\text{Fe}^{+3}(\text{CN})_6]^{-3}$ on the extreme right side.



✓ **Ligand:** The specie (atoms or group of atoms) that has lone pair of electrons and has an ability to donate it into vacant orbitals of central transition metal to form dative bond.

✓ **Coordination Number:** The number of dative bond present between metal ion and surrounding ligands.

Classification of Ligands:

On the basis of charged:

- 1) Negative Ligands:** which having negative charge. Such as Br^- , $\text{C}_2\text{O}_4^{2-}$, CO_3^{2-}
- 2) Neutral Ligands:** which having no charge on it. Such as H_2O^0 , $\text{NH}_2\text{-CH}_2\text{-CH}_2\text{-NH}_2^0$.
- 3) Positive Ligands:** which having positive charge on it. Like NO^+ (Nitrosonium).

On the basis of no. of electrons donated to central metal:

Monodentate Ligands:

- Those ligands which donate 01 lone pair of electrons to vacant d-orbitals of central metal.
- Such ligands may be negatively charged, or neutral.

Examples:

(i) Negatively charged: Br^- , OH^- , I^- , F^- , Cl^- , CN^- , and SCN^- .

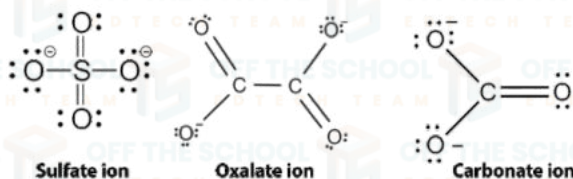
(ii) Neutral: H_2O^0 & NH_3^0 .

Bidentate Ligands:

Those ligands which have only two donatable electron pairs.

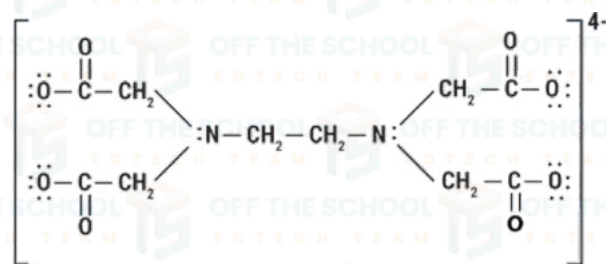
Examples: Sulphate ion (SO_4^{2-}), $\text{NH}_2\text{-CH}_2\text{-CH}_2\text{-NH}_2^0$, Oxalate ion ($\text{C}_2\text{O}_4^{2-}$)

Carbonate ion (CO_3^{2-}).



Polydentate or Multidentate Ligands:

Those ligands which donate more than 02 lone pair of electrons to vacant d-orbitals of central metal.



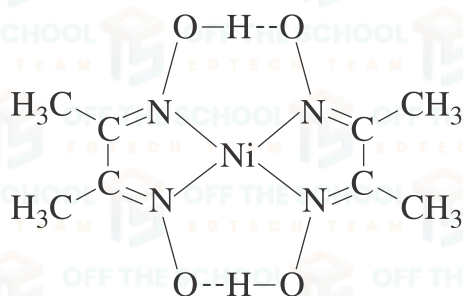
EDTA⁴⁻, Ethylenediaminetetraacetate ion

On the basis of Bonding:

Chelating agent: A bidentate or polydentate ligand which forms more than one dative bond to produce ring complexes called chelates.

Chelates: A complex of polydentate ligand with ring structure (resembles claw of crab).





Structure of nickel dimethylglyoxime. The overall structure is stabilized by hydrogen bonds.

LINK TO LEARNING

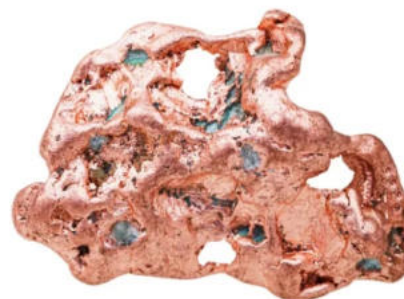
Do you think you understand naming coordination complexes? You can look over more examples and test yourself with online quizzes (<http://openstaxcollege.org/l/16namingcomps>) at the University of Sydney's site.

LINK 01



CHEMISTRY OF COPPER:

- It is the 9th member of 3d series.
- Its VSEC is $4s^1, 3d^{10}$.
- It is dense metal with reddish color.
- It is malleable and ductile.
- It has an excellent electrical conductivity & second to silver in conductivity series in its pure form.



METALLURGY OF COPPER:

Definition: Metallurgy is the branch of science and technology that deals with the study of metals, their properties, extraction from ores, purification, alloying, and various processes involved in their production and use.

Ore: All those minerals in which metals can be extracted easily and cheaply.

Extraction of Copper:

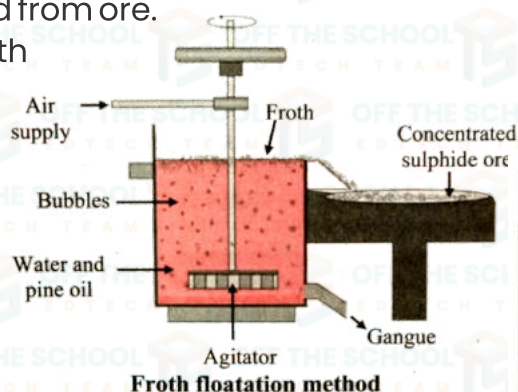
There are 06 steps involve during the extraction of copper from its ore, Cu is mainly extracted from its sulphide ore called "Chalcopyrite" or "Copper pyrite" (CuFeS_2) containing about 6% Cu.

1) Crushing or Grinding:

First, the copper ore is mined from the ground and then crushed into small pieces.

2) Concentration or Dressing of Ore:

- It is the process from which impurities are removed from ore.
- The impurities present in the ore is removed by Froth floatation process.
- In this process, ore is mixed with water-containing pine oil.
- Air is passed through it so that ore particles becomes coated with froth.
- Gangue particles settled down at bottom.
- The ore is concentrated to about 20–30%.





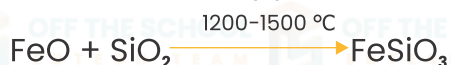
3) Roasting:

- Now, the concentrated copper ore is heated in a kiln or reverberatory furnace. This process is called roasting. It gets rid of any remaining impurities.



4) Smelting:

- It is the melting of ore at high temperature.
- The roasted ore is mixed with Silica (SiO_2) in order to remove impurities FeO as FeSiO_3 as a slag.
- The slag is removed from the surface of Copper matte (Cu_2S).



5) Bessemerization:

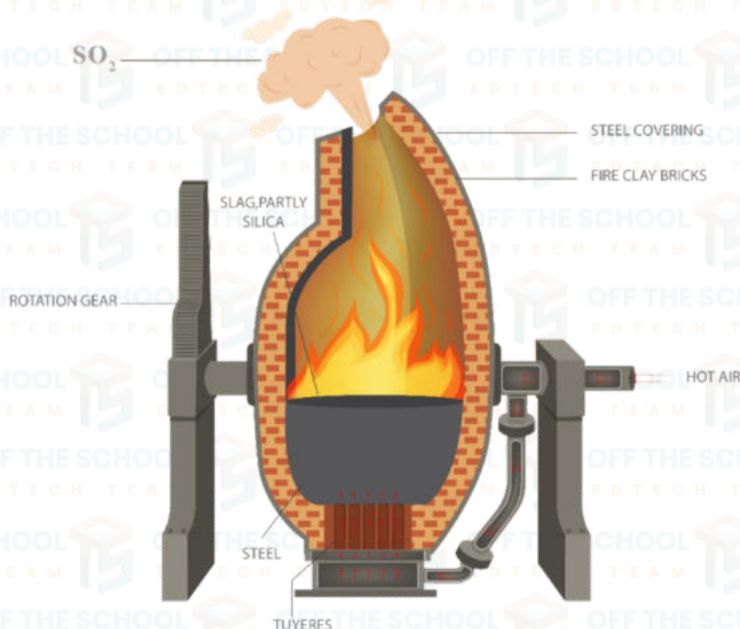
- The molten matte is introduced into a Bessemer converter.
- A blast of hot air is introduced into the converter from mid lower portion.
- The molten matte (Cu_2S) is further oxidized in this Bessemer converter.



OR

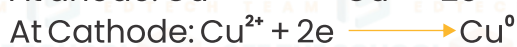


BESSEMER CONVERTER



6) Refining:

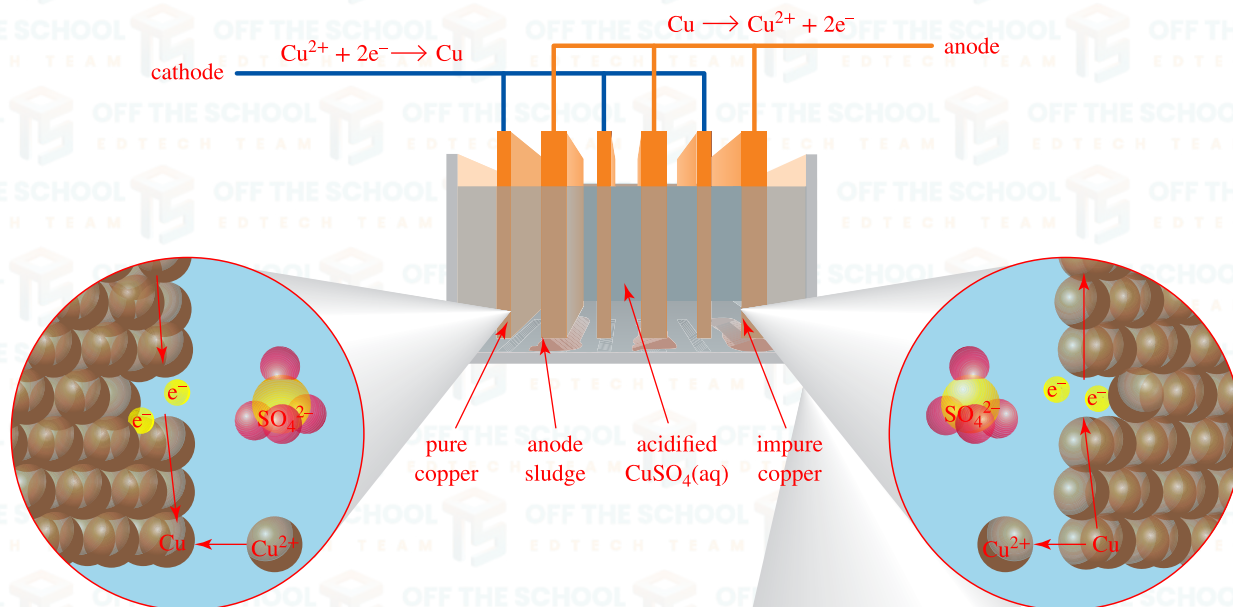
- Electrolytic refining is used to purify impure copper (blister Copper).
- It involves passing an electric current through acidified copper sulfate solution.
- Impure copper serves as the anode, and pure copper is deposited on the cathode.
- Impurities are removed as sludge or accumulate on the anode.

**LINK TO LEARNING**

Extraction of Copper online Video
(https://youtu.be/KBPv2p7Tlwo?si=ogQ6WmGBB5j9Vm_W).

LINK 01

Diagram of an electrolytic cell for the purification of copper



Rules for Naming Coordination Compounds

1. As with any ionic compound, the cation is named before the anion.
2. In naming a complex ion, the ligands are named before the metal ion.
3. In naming ligands, an 'o' is added to the root name of an anion. For example, the halides as ligands are called **fluoro**, **chloro**, **bromo**, and **iodo**; hydroxide is **hydroxo**; and cyanide is **cyano**. For a neutral ligand, the name of the molecule is used, with the exception of H_2O , NH_3 , CO , and NO .
4. The prefixes *mono-*, *di-*, *tri-*, *tetra-*, *penta-*, and *hexa-* are used to denote the number of simple ligands. The prefixes *bis-*, *tris-*, *tetrakis-*, and so on are also used, especially for more complicated ligands or ones that already contain *di-*, *tri-*, and so on.
5. The oxidation state of the central metal ion is designated by a Roman numeral in parentheses.
6. When more than one type of ligand is present, ligands are named in alphabetical order. Prefixes do not affect the order.
7. If the complex ion has a negative charge, the suffix *-ate* is added to the name of the metal. Sometimes the Latin name is used to identify the metal.

Some Common Ligands

Name	Structure
<i>Monodentate ligands</i>	
Ammonia	$\begin{array}{c} \text{H} \quad \ddot{\text{N}} \quad \text{H} \\ \\ \text{H} \end{array}$
Carbon monoxide	$:\text{C}\equiv\text{O}:$
Chloride ion	$:\ddot{\text{Cl}}:^-$
Cyanide ion	$[:\text{C}\equiv\text{N}:]^-$
Thiocyanate ion	$[:\ddot{\text{S}}-\text{C}\equiv\text{N}:]^-$
Water	$\begin{array}{c} \ddot{\text{O}} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{H} \end{array}$
<i>Bidentate ligands</i>	
Ethylenediamine	$\text{H}_2\ddot{\text{N}}-\text{CH}_2-\text{CH}_2-\ddot{\text{N}}\text{H}_2$
Oxalate ion	$\left[\begin{array}{cc} \text{:}\ddot{\text{O}}\text{:} & \text{:}\ddot{\text{O}}\text{:} \\ \diagdown & \diagup \\ & \text{C}=\text{C} \\ \diagup & \diagdown \\ \text{:}\ddot{\text{O}}\text{:} & \text{:}\ddot{\text{O}}\text{:} \end{array} \right]^{2-}$
<i>Polydentate ligand</i>	
Ethylenediaminetetraacetate ion (EDTA)	$\left[\begin{array}{cc} \text{:}\ddot{\text{O}}\text{:} & \text{:}\ddot{\text{O}}\text{:} \\ \parallel & \parallel \\ \text{:}\ddot{\text{O}}\text{:}-\text{C} & -\text{CH}_2- & \text{CH}_2-\text{C}-\text{:}\ddot{\text{O}}\text{:} \\ \diagdown & \diagup & \diagdown & \diagup \\ & \text{N}-\text{CH}_2-\text{CH}_2-\text{N} & \\ \diagup & \diagdown & \diagup & \diagdown \\ \text{:}\ddot{\text{O}}\text{:}-\text{C} & -\text{CH}_2- & \text{CH}_2-\text{C}-\text{:}\ddot{\text{O}}\text{:} \\ \parallel & \parallel \\ \text{:}\ddot{\text{O}}\text{:} & \text{:}\ddot{\text{O}}\text{:} \end{array} \right]^{4-}$



Determining the Oxidation Number of a Metal in a Complex



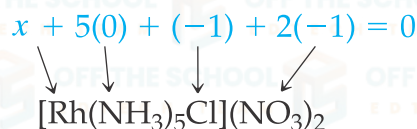
What is the oxidation number of the metal in $[\text{Rh}(\text{NH}_3)_5\text{Cl}](\text{NO}_3)_2$?

SOLUTION

Analyze We are given the chemical formula of a coordination compound and asked to determine the oxidation number of its metal atom.

Plan To determine the oxidation number of Rh, we need to figure out what charges are contributed by the other groups. The overall charge is zero, so the oxidation number of the metal must balance the charge due to the rest of the compound.

Solve The NO_3 group is the nitrate anion, which has a 1- charge. The NH_3 ligands carry zero charge, and the Cl is a coordinated chloride ion, which has a 1- charge. The sum of all the charges must be zero:



The oxidation number of rhodium, x , must therefore be +3.

Practice Exercise 1

In which of the following compounds does the transition-metal have the highest oxidation number?

- (a) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (b) $\text{K}_2[\text{PtCl}_6]$ (c) $\text{Rb}_3[\text{MoO}_3\text{F}_3]$
(d) $\text{Na}[\text{Ag}(\text{CN})_2]$ (e) $\text{K}_4[\text{Mn}(\text{CN})_6]$

Practice Exercise 2

What is the charge of the complex formed by a platinum(II) metal ion surrounded by two ammonia molecules and two bromide ions?

Practice Nomenclature

Naming Coordination Compounds

Give systematic names for the following compounds.

- (a) $[\text{Fe}(\text{CO})_5]$
(b) $\text{Na}_2[\text{Co}(\text{OH})_2(\text{OH})_4]$
(c) $[\text{Ag}(\text{NH}_3)_2]\text{Br}$
(d) $[\text{Cr}(\text{en})_3](\text{NO}_3)_3$
(e) $[\text{Pt}(\text{NH}_3)_4(\text{NO}_2)_2]\text{F}_2$
(f) $\text{K}_2[\text{Cu}(\text{CN})_4]$
(g) $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2](\text{NO}_3)_2$
(h) $\text{Na}[\text{Al}(\text{H}_2\text{O})_2(\text{OH})_4]$
(i) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2][\text{Cr}(\text{C}_2\text{O}_4)_2]$

Give systematic names for the following compounds.

- (a) $\text{Na}[\text{Au}(\text{CN})_2]$
(b) $[\text{Ni}(\text{CO})_4]$
(c) $[\text{CoCl}_6]^{3-}$
(d) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
(e) $\text{Na}_2[\text{Pt}(\text{CN})_4]$
(f) $\text{K}[\text{Cr}(\text{NH}_3)_2(\text{OH})_2\text{Cl}_2]$
(g) $[\text{Mo}(\text{NCS})_2(\text{en})_2]\text{ClO}_4$
(h) $[\text{Mn}(\text{NH}_3)_2(\text{OH})_2(\text{OH})]\text{SO}_4$
(i) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{I}$

Give systematic names for the following compounds.

- (a) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
(b) $[\text{Fe}(\text{en})_3]\text{PO}_4$
(c) $[\text{Co}(\text{NH}_3)_6]\text{SO}_4$
(d) $[\text{Co}(\text{NH}_3)_6]_2(\text{SO}_4)_3$
(e) $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_2$
(f) $(\text{NH}_4)_2[\text{PtCl}_4]$
(g) $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{NO}$

Give systematic names for the following compounds.

- (a) $\text{K}_2[\text{PdCl}_6]$
(b) $(\text{NH}_4)_2[\text{PtCl}_6]$
(c) $[\text{Co}(\text{en})_3]\text{Cl}_2$
(d) $[\text{Co}(\text{en})_3]\text{Cl}_3$
(e) $[\text{Cr}(\text{en})_2(\text{NH}_3)_2](\text{NO}_3)_2$
(f) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
(g) $(\text{NH}_4)_3[\text{CuCl}_5]$

Write formulas for the following.

- (a) diamminedichlorozinc
(b) tin(IV) hexacyanoferrate(II)
(c) tetracyanoplatinate(II) ion
(d) potassium hexachlorostannate(IV)
(e) tetraammineplatinum(II) ion
(f) hexaamminenickel(II) bromide
(g) tetraamminecopper(II) pentacyanohydroxoferrate(III)
(h) diaquacyanocopper(II)
(i) potassium hexachloropalladate(IV)

Write formulas for the following compounds.

- (a) *trans*-diamminedinitroplatinum(II)
(b) rubidium tetracyanozincate
(c) triaqua-*cis*-dibromochlorochromium(III)
(d) pentacarbonyliron(0)
(e) sodium pentacyanocobaltate(II)
(f) hexammineruthenium(III) tetrachloronickelate(II)
(g) sodium tetracyanocadmiate
(h) hexaamminecobalt(III) chloride

